PROBLEM STATEMENT CODE: PS000109 • DIGITAL OCCUPATIONAL HEALTH MONITORING

WorkCheck Kachchh (કચ્છ શ્રમિક સુરક્ષા)

CORE MISSION: Find exposed informal workers via **public health clinics**, trigger check-ups via **₹2,015 solar sensors**, and track referrals i

Worker Health Pass.

STEP 1 • SCREEN	STEP 2 • SENSE	STEP 3 • EXAMINE	STE
1-Min Clinic Check Past Jobs & Symptoms	**₹2,015 Solar Node** Heat (WBGT) & Dust Alert	**BOCW Mobile Van** Free Doctor Check-Up	**e-Shram** Portable

TEAM ID	TRACK CATEGORY	INSTITUTION / ORGANIZATION	TEAM LEADER
TM005024	College Track (Engineering)	LDRP Institute of Technology & Research, Gandhinagar, Gujarat	Jay Patel (Tean

CHALLENGE DEFINITION

Problem Statement & Real-World Context

Focus: Roo

Root-Cause Flowchart & Field Evidence

1.8 LAKH

1. Unregistered Informal Laborers
Salt pans, quarries, ports & cement off factory rolls

▼

AGE 45+/40+

2. Outdated Age-Only Check-Up Rules
Rule 9 & 5 miss younger workers harmed by daily dust/heat

▼

87.5% HEAT

3. Extreme Heat & Silica Dust Harm
2.1× kidney risk • 26% > silica limit • 2,067 hot hrs (Kandla)

▼

15% AUDIT

4. Unrecorded Disease & Wage Fear
182 silicosis cases & 80 deaths (PTRC) vs 562 logged nationally

4 Target Groups & Why Market Fails

① Agariya Salt Rakers
Extreme Sun & Brine Glare

② Stone Quarry
Respirable Silica L

③ Port & Dock Loaders
Kandla & Mundra Marine Heat

④ Mobile Van L
BOCW Dhanvanta

WHY EXISTING MARKET SOLUTIONS FAIL INFORMAL LA

₹4,100 / Mo

US Wearable Patches (Kenzen, Slate
560× too costly for daily-wage workers

₹5.5 Lakhs

Lab Dust Monitors (TSI DustTrak)
Measures air only — zero link to doctor refe

Payroll Only

Corporate EHS Software (OccuCare
Tracks factory staff — misses migrating lab

SOLUTION APPROACH •

Proposed Solution & Innovation Novelty

Ethos: F

END-TO-END WORKFLOW DIAGRAM • WORKS 100% EVEN WITHOUT HARDWARE (SENSOR-OPTIONAL)

1. PRIORITIZE	2. 1-MIN CHECK	3. 3-RULE TRIAGE	4. VAN EXAM	5.
Heat & Dust Map Rank High-Risk Sites	**Clinic / Volunteer** Log Job Years & Symptoms	**Auto Risk Engine** Age • Past Dust • Symptoms	**BOCW Mobile Van** Doctor X-Ray & Creatinine	**QR** Portab

Core Novelty

Portable QR Health Pass
12-Digit e-Shram UAN + ABHA ID follows worker

Past Exposure Memory
Counts career dust/heat years (e.g., 20 yrs at crusher)

100% DPDP Act Privacy
Worker OTP lock • Anonymous DDO taluka heatmaps

Technical Moat

150mm Black Copper Globe
True ISO 7243 solar radiant heat (zero wind error)

Coastal Humidity Math Fix
Prevents sea-air false dust spikes ($R^2 > 0.88$)

1-Click ABDM FHIR Sync
Instant HL7 FHIR R4 report deposit to ABHA

"Seva First" Va

< ₹7.30 / Worker / Month
₹0 worker cost — 560× cheaper th

100% Cess & DMF Funded
Backed by Gujarat BOCW 1% Ce

Gujarati • Hindi • English
Native trilingual UI (ગુજરાતી) + R

LIVE SANDBOX & DEMO

Interactive 4-Persona Working Prototype (DDO, Van Doctor, Supervisor & Worker Views):
https://jaypatel29042008-glitch.github.io/WorkCheck-Kachchh-SSIP/

SYSTEM ARCHITECTURE •

Technical Architecture || Methodology

① EDGE SENSING LAYER
Worksite Hardware (Teal)

SOLAR IOT NODE	**₹2,015 Bulk BOM • ESP32** Installed at salt pans, quarries & ports
150mm BLACK GLOBE	**True Radiant Heat (Tg)** Eliminates wind error ($R^2 = 0.92$)
DUST + CLIMATE + UV	**PMS5003 • BME280 • UV** Laser PM2.5/PM10, Temp/RH & Glare
RGB FLAG LAMP	**4-Color Work-Rest Light** Green / Yellow / Orange / Red on site
ZERO-SENSOR MODE	**100% Software Fallback** Clinic screening works without hardware

② COMPUTATION & RULES
FastAPI + SQL Engine (Orange)

ISO 7243 WBGT MATH	**0.7 Tnw + 0.2 Tg + 0.1 Ta** ACGIH heavy-work heat strain index
COASTAL DUST FIX	**PM_raw / [1+0.25·RH²/(1−RH)]** Prevents sea humidity fogging
10-TABLE SQL DB	**30-Day Raw + Rollups** Longitudinal worker exposure history
3 ROUTING RULES	**1. Age 40/45+ • 2. Quartz Yrs** 3. Symptoms (Dark urine, cough)
DOCTOR OWNERSHIP	**Clinician-Reviewed Rules** Software flags risk; van doctors diagnose

③ DELIVERY & GO
Worker, Van & DDO

WORKER QR PASS	**12-Digit e-S** Portable bilir contractors
BOCW MOBILE VAN	**Dhanvanta** Spirometry,
1-CLICK ABHA SYNC	**HL7 FHIR** Deposits repo Locker
AUTO FORM 5 CERT	**Gujarat OS** Annual statut
DDO GIS HEATMAP	**Anonymou** 1-Click BOC CSVs

ENGINEERING STACK •

Technology Stack & Feasibility

Component

A. EMBEDDED IOT & HARDWARE

TRL-5 • VALIDATED

ESP32 Dual-Core
240MHz + Offline Flash Buffer

BME280 + PMS5003
Laser Dust, Temp/RH & UV

150mm Black Globe
Copper Float + 10k Thermistor

₹2,015 Unit BOM
5V Solar + Li-ion + IP65 Box

B. BACKEND & RULES ENGINE

TR

Python 3.10 + FastAPI
Async REST + OpenAPI Docs

10-Table ANSI S
30-Day Raw + Car

ISO 7243 + Aerosol
WBGT & Humidity Correction

21/21 Tests Passi
Automated Integrat

C. FRONTEND & CLINICAL UI

TRL-6 • LIVE PWA

Offline-First PWA
Mobile-First Tailwind CSS

Leaflet GIS + Chart.js
Live Taluka Heatmaps

Gujarati • Hindi • EN
Native Trilingual UI (ગુજરાતી)

4 User Personas
Worker, Site, Van Doctor, DDO

D. STANDARDS & PRIVACY

TR

ABDM HL7 FHIR R4
1-Click DiagnosticReport Sync

12-Digit e-Shram
National QR Health

10 Public Datasets
ERA5, CAMS, CPCB, MSHA, OSHA

100% DPDP Act
Worker OTP + And

IMPACT ASSESSMENT • SIH SECTION 05

Socio-Economic Impact & Viability

Qua

1. CLINICAL IMPACT

>90% Follow-Up
2.1× Kidney Risk Addressed

KIDNEY PROTECTION

Prevents AKI & CKDu
87.5% salt-pan heat stress & 2.1× kidney risk

SILICOSIS CAPTURE

Early Lung Screening
Flags quartz dust years before lung fibrosis

ZERO LEAKAGE

>90% Referral Tracking
Follows every case to clinic vs 15% audit

2. FRUGAL ECONOMICS

₹7.30 / Month
560× Cheaper Than Wearables

560× CHEAPER

₹7.30 vs ₹4,100 / Mo
Replaces costly US patches (Kenzen)

₹2,015 SITE NODE

Replaces ₹5.5L DustTrak
1 solar node covers entire worksite

₹2.97L PILOT

50 Sites • 1,000 Workers
Year-1 district budget (₹88/worker/yr)

3. PRODUCTIVITY

~358 Hrs/Yr
~20% Severe Heat Avoided

6 AM – 3 PM SHIFT

Cool-Shift Optimization
Avoids ~20% peak heat (~358 hrs/yr)

RGB REST LAMPS

Zero Employer Disputes
Objective ISO 7243 work-rest light

AMENITY AUDIT

Water & Shade Tracking
Verifies drinking water & rest tents

4. GO

₹0 W
100% BOCW

10

₹0 Paid by Lal
Health data is ne

CESS

BOCW 1% Ce
Uses existing sta

RULE

Auto Form 5 &
Instant complian

EXECUTION TIMELINE • SIH SECTION 06

Implementation Roadmap & Milestones

From

STAGE 1: LAB VALIDATION

Oct – Nov 2026 • TRL-5
STATUS: COMPLETED

1. Datasets Analyzed
ERA5, CAMS, CPCB & OSHA (5 Sites)

↓

2. Core Engine Built
10-Table SQL + ISO 7243 Rules

↓

3. 4-Persona Web App
1-Click ABHA FHIR Publishing

↓

4. Verified & Live
21/21 Tests • GitHub Pages Live

STAGE 2: PILOT FIELD TRIAL

Nov – Dec 2026 • TRL-6
STATUS: PLANNED

1. Field Site Nodes
2–3 Solar Nodes (Rapar & Abdasa)

↓

2. Lab Calibration
Validate vs Reference Instruments

↓

3. PHC / CHC Pilot
1-Min Exposure Check at 1 Clinic

↓

4. Clinical Sign-Off
CDHO Kachchh & NIOH Review

STAGE 3: VAN INTEGRATION

Jan – Mar 2027 • TRL-7
STATUS: PLANNED

1. Mobile Van Fleet
BOCW Dhanvantari Rath Link

↓

2. 50-Site Rollout
Cover 1,000 Informal Workers

↓

3. QR Health Passes
Bilingual e-Shram PVC Cards

↓

4. Live ABHA Sync
Spirometry & Creatinine Deposits

STAGE 4: I

Apr 2
STAT

1. All 10 Taluk
Scale to 10,000+

↓

2. ESIC Dispe
Bhuj & Gandhid

↓

3. SEWA Insu
Parametric Heat

↓

4. District Poli
Anonymized DD

SQUAD CAPABILITIES

Multi-Disciplinary Team & Mentorship

TL

Jay Patel

Team Leader & Architect

- **3-Layer Architecture**
  End-to-End Platform Design
- **10-Table SQL & Rules**
  ISO 7243 & Aerosol Engine
- **ABDM HL7 FHIR R4**
  1-Click ABHA Locker Sync
- **21/21 Test Harness**
  Automated CI Verification

M2

Hardware & IoT Lead

Embedded & Calibration

- **ESP32 Dual-Core**
  Firmware & Wokwi Model
- **150mm Black Globe**
  Radiant Heat Calibration
- **Sensor Array Wiring**
  PMS5003, BME280 & UV
- **₹2,015 Solar BOM**
  5V Solar + Li-ion Design

M3

Public Health Lead

Field Ops & Localization

- **12-Digit e-Shram QR**
  Portable Worker Health Pass
- **BOCW Mobile Van**
  Dhanvantari Rath Liaison
- **Gujarati • Hindi • EN**
  Trilingual UI (ગુજરાતી)
- **1-Min Clinic Triage**
  ASHA & ANM Workflow

Comp

Hygiene

- **Gujarat OSH**
  Auto Form 5 Ce
- **Factories Act 4**
  Central Rule 5 M
- **DPDP Act 202**
  OTP Lock & An
- **10 Public Data**
  Empirical Hazar

Faculty Mentor: Prof. / Dr. Faculty Mentor

Department of Computer Engineering / IT, LDRP-ITR, Gandhinagar

Ready for Ju